QCU ACADEMIC OS · AY 2026–2027

WS 101

Web Systems and Technologies 1 — Complete Study Reference

Course: WS 101 | Units: 3 units | Year Level: 1st Year | Type: HTML/CSS/JS + PHP/MySQL + Laravel

CHAPTER 00

Course Overview & How the Web Works

WS 101 starts by teaching you to build the **front-end of the web** — the structure (HTML) and visual presentation (CSS) of every website you have ever used — then continues into **server-side scripting**: PHP, MySQL, and the Laravel framework. Unlike CC 101's pure theory, this is hands-on: every concept here is something you will type and immediately see rendered. This guide takes that further — wherever possible, you will see **live, actually-rendered output** next to the code that builds it, not just a screenshot.

> **Critical Path — Layer 1 Foundation**
>
> WS 101 unlocks **WS 102** (Web Systems 2, Sem 2) directly, and indirectly feeds **HCI 101** (Human Computer Interaction, Year 2) and **IPT 102** (Integrative Programming 2, Year 3) which both require WS 101 + HCI 101 as prerequisites. Everything visual you build in your entire BSIT degree traces back to this course.

The Client-Server Model

Every website you visit involves two computers talking to each other: a **client** (your phone/laptop, running a browser) and a **server** (a remote

computer storing the website's files). This conversation happens in a clear request-response pattern.

1. **You type a URL or tap a link**
   e.g. `https://qcu.edu.ph`
2. **Browser sends an HTTP Request**
   A formatted message asking the server: "send me the file at this address." DNS first resolves the domain name to an IP address (covered in CC 101 Ch. 7).
3. **Server processes the request**
   The server locates the requested HTML file (or generates it dynamically) and prepares a response.
4. **Server sends an HTTP Response**
   Contains a status code (200 OK, 404 Not Found, 500 Server Error) plus the actual HTML, CSS, JS, and asset files.
5. **Browser renders the page**
   Parses HTML into a DOM tree, parses CSS into a CSSOM tree, combines them into a Render Tree, then paints pixels to your screen. This entire pipeline happens in milliseconds.

How a Browser Renders a Page — The Pipeline

Stage	What Happens
1. Parse HTML → DOM	Browser reads your HTML top to bottom and builds the **DOM** (Document Object Model) — a tree structure representing every element and its nesting relationship.
2. Parse CSS → CSSOM	Browser reads all CSS rules and builds the **CSSOM** (CSS Object Model) — a tree of every style rule and what it applies to.
3. Combine → Render Tree	DOM + CSSOM are merged into a Render Tree containing only visible elements with their final computed styles.
4. Layout (Reflow)	Browser calculates the exact size and position of every element on the page based on the Render Tree.
5. Paint	

Stage	What Happens
	Browser fills in actual pixels — colors, text, images, borders — based on the Layout step.

Filipino Analogy — Client-Server Model

Think of ordering at a **carinderia counter**. You (client) place an order (HTTP Request) for "1 Adobo with rice" (a specific URL). The kitchen (server) prepares it and hands it back (HTTP Response) — either your food (200 OK) or "ubos na po" / sold out (404 Not Found). You then plate it yourself at your table exactly how you like it (the browser rendering the HTML/CSS into a visual page).

HTML vs CSS vs JavaScript — The Three Pillars

Layer	Job	Analogy	Covered In
HTML	Structure & content — the skeleton	The walls, rooms, and frame of a house	WS 101 (this course)
CSS	Presentation & styling — the appearance	Paint, furniture, interior design	WS 101 (this course)
JavaScript	Behavior & interactivity — the logic	Electricity, plumbing, automated doors	WS 101 (this course, Weeks 5–6) — then deepened in WS 102

CHAPTER 01

HTML Document Structure & Semantic Elements

HTML (HyperText Markup Language) is not a programming language — it is a **markup language**. It does not compute logic; it describes the structure and

meaning of content using tags. Every HTML document follows the same basic skeleton.

The Minimum HTML5 Boilerplate

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My First Page</title>
</head>
<body>
  <h1>Hello, QCU!</h1>
</body>
</html>
```

Line-by-Line Breakdown

Line	Purpose
`<!DOCTYPE html>`	Tells the browser this is an HTML5 document. Must be the very first line. Without it, browsers fall into "quirks mode" with inconsistent, legacy rendering behavior.
`<html lang="en">`	The root element wrapping all content. The `lang` attribute helps screen readers and search engines identify the page language.
`<head>`	Contains metadata — information ABOUT the page that is NOT directly displayed (title, character encoding, linked CSS/fonts, viewport settings).
`<meta charset="UTF-8">`	Sets character encoding so special characters (ñ, emoji, ₱) display correctly. UTF-8 is the universal standard — connects directly to Unicode from CC 101 Ch. 4.
`<meta name="viewport"...>`	Controls how the page scales on mobile devices. Without this, your page would render

Line	Purpose
	zoomed-out and tiny on a phone screen like your GT 30 Pro.
`<title>`	Text shown in the browser tab and search engine results. Not visible on the page itself.
`<body>`	Contains ALL visible content — everything the user actually sees and interacts with.

Semantic vs Non-Semantic Elements

Semantic elements clearly describe their meaning to both the browser and the developer. **Non-semantic** elements (`div` , `span`) give no information about their content — they are generic containers.

Semantic Tag	Meaning	Typical Use
`<header>`	Introductory content	Site logo, title, main navigation
`<nav>`	Navigation links	Menu bar, links to other pages/sections
`<main>`	The dominant, unique content of the page	Only ONE per page — the core content
`<section>`	A thematic grouping of related content	"About Us" section, "Services" section
`<article>`	Self-contained, independently distributable content	A blog post, a news story, a forum post
`<aside>`	Content tangentially related to the main content	Sidebar, related links, advertisement
`<footer>`	Closing content	Copyright, contact info, social links

Live: Semantic Page Skeleton

This is what a semantic layout actually looks like rendered — each labeled box is a real semantic element, not a generic `div` .

LIVE PREVIEW

Why Semantic HTML Matters

Screen readers (accessibility tools for visually impaired users) rely on semantic tags to navigate a page intelligently. Search engines (Google) also weigh semantic structure when ranking pages. A page built entirely from generic `<div>` tags works visually but is invisible in meaning to machines.

The HTML Document Outline (Element Nesting)

Concept	Rule
Parent / Child	An element written inside another is its "child." The outer one is the "parent." `<body>` is the parent of everything visible.
Nesting must be proper	Tags must close in the reverse order they opened: `<p><strong>text</strong></p>` is correct. `<p><strong>text</p></strong>` is broken/invalid.
Block-level elements	Start on a new line, take full available width by default. Examples: `div, p, h1-h6, section, header, footer, ul, table`.
Inline elements	Do not start a new line, only take up as much width as their content needs. Examples: `span, a, strong, em, img, label`.

Forward Link — CC 102 / PF 101

The parent-child nesting concept you are learning right now is the exact same hierarchical thinking you will use for class inheritance in **PF 101** (Object-Oriented Programming, Year 2) — a child class inherits structure from its parent class, just like a nested HTML element inherits positioning context from its parent.

Text, Lists, Links & Images

These are the elements that make up the actual readable content of nearly every web page — headings, paragraphs, text emphasis, lists, hyperlinks, and images.

Headings & Paragraphs

HTML provides 6 levels of headings, `<h1>` through `<h6>`, in decreasing importance and size. There should only be **one** `<h1>` **per page** — it represents the main topic.

HTML

```
<h1>Main Title</h1>
<h2>Section Title</h2>
<h3>Subsection</h3>
<p>This is a normal
paragraph of text.</p>
<p><strong>Bold</strong> and
<em>italic</em> text.</p>
<blockquote>A quoted
passage.</blockquote>
```

LIVE PREVIEW

Text Formatting Tags Reference

Tag	Renders As	Semantic Meaning
`<strong>`	Bold	Strong importance (screen readers emphasize vocally)
`<b>`	Bold	Visual bold only, no added importance — prefer `<strong>`
`<em>`	Italic	Emphasized stress — changes meaning of a sentence

Tag	Renders As	Semantic Meaning
`<i>`	Italic	Visually offset text (foreign words, technical terms) — no emphasis
`<mark>`	Highlighted	Marked/highlighted relevance, like a highlighter pen
`<small>`	Smaller text	Fine print, disclaimers
`<del>`	Strikethrough	Deleted/removed text
`<code>`	Monospace	Inline code snippet
` `	Line break	Forces a new line WITHOUT starting a new paragraph (self-closing, no closing tag)
`<hr>`	Horizontal line	Thematic break between content sections (self-closing)
`<tt>`	Monospace	Deprecated — HTML5 removed it. Use `<code>` instead. Still shown on lecture slides as "Monospaced Text."
`<big>`	Larger text	Deprecated — removed in HTML5. Use CSS `font-size` instead.
`<font>`	Sets face/color/size	Deprecated — mixed styling into content. Use CSS instead: `font-family`, `color`, `font-size` properties replace its `face`, `color`, `size` attributes.
`<center>`	Centers content	Deprecated — use CSS `text-align: center` (text) or `margin: 0 auto` (block elements) instead.

Exam-Relevant but Obsolete — Physical vs Logical Style

Older HTML teaching material (including your lecture deck) splits text formatting into two categories: **Physical Style** — tags that dictate exact visual appearance regardless of meaning (`<b>` , `<i>` , `<tt>` , `<u>` , `<s>` , `<big>` , `<small>` , `<sub>` , `<sup>`) — versus **Logical Style** — tags that describe meaning and let the browser decide the visual (`<em>` , `<strong>` , `<cite>`). This terminology may appear in a quiz. In modern practice, the "logical" category is exactly what the **Semantic Meaning** column above is teaching you — `<strong>` / `<em>` over `<b>` / `<i>` for the same reason.

Lists

HTML

```
<h3>Unordered</h3>
<ul>
  <li>HTML</li>
  <li>CSS</li>
</ul>

<h3>Ordered</h3>
<ol>
  <li>Plan</li>
  <li>Build</li>
</ol>

<h3>Description</h3>
<dl>
  <dt>HTML</dt>
  <dd>Structure</dd>
</dl>
```

LIVE PREVIEW

Links — The Anchor Tag

Attribute	Purpose	Example
`href`	Destination URL — required attribute	

Attribute	Purpose	Example
		`href="https://qcu.edu.ph"`
`target="_blank"`	Opens link in a new browser tab	`target="_blank"`
`rel="noopener"`	Security best practice paired with target="_blank" — prevents the new tab from accessing your original page	`rel="noopener noreferrer"`

Absolute vs Relative Paths

Type	Example	When to Use
Absolute	`https://google.com`	Linking to an external website
Relative	`about.html` or `../images/photo.jpg`	Linking to another file within your own project — works regardless of domain, essential for local testing in VS Code Live Server

Images

The `<img>` tag is **self-closing** (no separate closing tag) and requires two key attributes: `src` (the image file path) and `alt` (alternative text describing the image for accessibility and for when the image fails to load).

HTML

```
<figure>
  <img
    src="landscape.jpg"
    alt="Mountains at
         sunrise"
    width="300">
  <figcaption>
    A scenic view
```

LIVE PREVIEW

```
  </figcaption>
</figure>
```

Gotcha Alert — Always Include alt Text

If you forget `alt`, screen readers announce the raw filename to visually impaired users — useless and unprofessional. If an image fails to load (broken link), the alt text displays in its place, telling the user what was supposed to be there.

CHAPTER 03

Tables & Forms

Tables display tabular (grid-based) data. Forms collect input from users — the entry point for nearly every interactive feature on the web, from login pages to GCash transfers.

Tables

Tag	Purpose
`<table>`	The container for the entire table
`<thead>`	Groups the header row(s)
`<tbody>`	Groups the main data rows
`<tr>`	Table Row
`<th>`	Table Header cell — bold, centered by default
`<td>`	Table Data cell — regular content cell
`colspan`	Attribute — makes a cell span multiple columns
`rowspan`	Attribute — makes a cell span multiple rows

HTML

```
<table>
  <thead>
    <tr>
      <th>Subject</th>
      <th>Units</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>CC 101</td>
      <td>3</td>
    </tr>
    <tr>
      <td>WS 101</td>
      <td>3</td>
    </tr>
  </tbody>
</table>
```

LIVE PREVIEW

Forms

The `<form>` tag wraps all input controls. Its `action` attribute specifies where submitted data goes; its `method` attribute specifies how (`GET` appends data visibly to the URL, `POST` sends it invisibly in the request body — always use POST for passwords/sensitive data).

Common Input Types

Type	Renders As	Use Case
`text`	Single-line text box	Name, username
`email`	Text box with email validation	Email address — browser checks for @ format
`password`	Text box that masks input (••••)	Passwords

Type	Renders As	Use Case
`number`	Numeric stepper input	Age, quantity
`checkbox`	Square box, multiple selectable	Agree to terms, multi-select options
`radio`	Circle button, ONLY one selectable per group (same `name` attribute)	Gender, single-choice selection
`date`	Calendar picker	Birthdate
`submit`	Clickable button that sends the form	"Submit" / "Login" button
`file`	File upload chooser	Profile picture, document upload

Live: A Complete Registration Form

HTML

```
<form>
  <label for="nm">Name</label>
  <input type="text" id="nm"
    placeholder="Juan Dela Cruz">

  <label for="em">Email</label>
  <input type="email" id="em">

  <label>Section</label>
  <select>
    <option>SBIT1D</option>
    <option>SBIT1E</option>
  </select>

  <label>
    <input type="checkbox">
    I agree to terms
  </label>

  <button type="submit">
    Register
```

LIVE PREVIEW — TAPPABLE

```
  </button>
</form>
```

Always Pair label with for/id

The `<label for="nm">` attribute must match the input's `id="nm"` exactly. This connects them so clicking the label text also focuses/activates the input — critical for accessibility and a noticeably better mobile tapping experience.

Validation Attributes

Attribute	Effect
`required`	Form cannot submit until this field has a value
`minlength` / `maxlength`	Sets minimum/maximum character count
`min` / `max`	Sets minimum/maximum numeric value (for type="number")
`pattern`	Custom regex pattern the input must match
`placeholder`	Faded hint text shown when the field is empty — NOT a substitute for a real `<label>`

CHAPTER 04

Multimedia, Entities & Global Attributes

This chapter rounds out HTML with embedding audio/video, special characters, and the universal attributes available on every HTML element.

Audio & Video

HTML

```
<video controls width="320">
  <source src="clip.mp4" type="video/mp4">
  Your browser does not support video.
</video>

<audio controls>
  <source src="song.mp3" type="audio/mpeg">
</audio>
```

Attribute	Effect
`controls`	Shows play/pause/volume/seek UI controls
`autoplay`	Starts playing automatically on page load (browsers often block this for sound; use with `muted`)
`loop`	Restarts playback automatically when it ends
`muted`	Starts with sound off
`poster`	(video only) Image shown before playback starts

The iframe Element

An `<iframe>` embeds an entirely separate HTML document inside the current page — like a window into another website. This is literally the technique used to build every "live preview" box throughout this guide.

HTML

```
<iframe
  src="https://www.youtube.com/embed/VIDEO_ID"
```

```
  width="320" height="180">
</iframe>
```

Frames & Frameset — Legacy Predecessor to iframe

Before `<iframe>` existed, HTML used `<frameset>` to split the ENTIRE browser window into multiple independent panels, each loading a separate HTML file. Your lecture deck still teaches this.

HTML

```
<html>
<head><title>HTML Frames</title></head>
<frameset rows="10%,80%,10%">
  <frame name="top" src="AboutUs.html">
  <frame name="main" src="Gallery.html">
  <frame name="bottom" src="ContactUs.html">
</frameset>
</html>
```

Tag/Attribute	Purpose
`<frameset>`	Replaces `<body>` entirely — splits the window using `rows` or `cols` (e.g. `rows="10%,80%,10%"`)
`<frame />`	One panel inside a frameset — `src` points to the HTML file it loads, `name` lets links target it
`<noframes>`	Fallback content for browsers that don't support frames

Deprecated — Do Not Use in Real Projects

`<frameset>` is obsolete in HTML5: it breaks bookmarking, browser back/forward, SEO, and accessibility, since the URL never reflects which content is showing. `<iframe>` (above) replaced it — it embeds one document inside a normal page instead of replacing the whole page. Know `<frameset>` / `<frame>` / `<noframes>` for the exam; use `<iframe>`, or better, CSS layout, for actual work.

HTML Entities — Special Characters

Some characters have special meaning in HTML (like `<` and `>` for tags) and cannot be typed directly into content. **Entities** are escape codes that represent them safely.

HTML

```
&lt;  &gt;  &amp;
&copy;  &hearts;
   &quot;
```

LIVE PREVIEW

Entity Code	Renders	Meaning
`<`	<	Less than
`>`	>	Greater than
`&`	&	Ampersand
`&quot;`	"	Quotation mark
` `	(space)	Non-breaking space — prevents line wrap at that point
`&copy;`	©	Copyright symbol
`&#8369;`	₱	Philippine Peso sign (numeric entity)

Global Attributes

These attributes can be applied to virtually ANY HTML element, regardless of tag type.

Attribute	Purpose
`id`	A UNIQUE identifier — only one element per page may have a given id. Used for CSS

Attribute	Purpose
	targeting, JS selection, and anchor links (`#overview`).
`class`	A reusable label — MANY elements can share the same class. The primary way to target groups of elements with CSS.
`style`	Inline CSS applied directly to that one element. Convenient but considered poor practice for real projects (mixes structure with presentation).
`title`	Tooltip text shown on mouse hover.
`data-*`	Custom data attributes for storing extra information accessible via JavaScript, e.g. `data-student-id="12-3456"`.
`hidden`	Hides the element completely from the rendered page.

id vs class — The Most Important Distinction

	`id`	`class`
Uniqueness	ONE per page	Reusable on MANY elements
CSS Selector	`#myId { }`	`.myClass { }`
Specificity	Higher (overrides class rules)	Lower
Best for	One specific unique element (a particular header, a single button)	A reusable style applied to many elements (all "card" components)

HTML Comments

HTML

```
<!-- This is a comment. It is invisible
     to the user but visible in source code. -->
<p>Visible text</p>
```

Comments are useful for leaving notes for yourself or teammates, and for temporarily disabling a block of code without deleting it.

CHAPTER 05

CSS Fundamentals

CSS (Cascading Style Sheets) controls how HTML elements LOOK — colors, spacing, fonts, layout. The "Cascading" in the name refers to the rules that determine which style wins when multiple rules target the same element.

History of CSS (exam trivia)

CSS was first proposed by **Håkon Wium Lie** on October 10, 1994. The first W3C CSS Recommendation (**CSS1**) was released in 1996, with **Bert Bos** as co-author — he's regarded as CSS's co-creator alongside Lie.

Three Ways to Add CSS

Method	Where It Lives	Example	Best Practice?
Inline	Directly on the element via `style` attribute	`<p style="color:red">`	Avoid — highest specificity, hard to maintain, mixes concerns
Internal	Inside a `<style>` tag in the document `<head>`	`<style>p{color:red}</style>`	OK for small single-page demos
External	A separate `.css` file linked via `<link>`	`<link rel="stylesheet"`	Best practice — reusable across multiple pages, clean separation

Method	Where It Lives	Example	Best Practice?
		`href="style.css ">`	

CSS Syntax Anatomy

```css
selector {
  property: value;
  property: value;
}

/* Real example: */
h1 {
  color: #3B82F6;
  font-size: 28px;
}
```

Part	Role
Selector	WHICH element(s) to style (`h1` in the example)
Declaration Block	Everything inside the curly braces `{ }`
Property	WHAT aspect to change (`color` , `font-size`)
Value	The setting applied to that property (`#3B82F6` , `28px`)
Declaration	One full property:value; pair

CSS Selectors — Full Reference

Selector	Syntax	Targets
Universal	`*`	EVERY element on the page

Selector	Syntax	Targets
Element / Type	`p`	Every `<p>` tag
Class	`.card`	Every element with `class="card"`
ID	`#header`	The one element with `id="header"`
Descendant	`nav a`	Any `<a>` nested ANYWHERE inside a `<nav>`
Child	`nav > a`	Only `<a>` tags that are DIRECT children of `<nav>`
Grouping	`h1, h2, h3`	Applies the same rule to all listed selectors
Attribute	`input[type="text"]`	Elements with a matching attribute value
Pseudo-class	`a:hover`	An element in a specific STATE (hovered, focused, first child)
Pseudo-element	`p::first-line`	A specific PART of an element's content

Common Pseudo-Classes

Pseudo-class	Triggers When
`:hover`	Mouse pointer is over the element (desktop only — no true hover on touchscreens)
`:focus`	Element is currently selected/active (e.g. a clicked/tapped input field)
`:active`	Element is being clicked/tapped at this exact moment
`:first-child`	Element is the first child of its parent
`:last-child`	Element is the last child of its parent

Pseudo-class	Triggers When
`:nth-child(n)`	Element is the nth child — supports formulas like `:nth-child(2n)` for every even item

Live: Selector Specificity in Action

Same `<p>` tag, three competing rules. Watch which one wins.

CSS

```
p { color: gray; }       /*
weakest */

.note { color: orange; } /*
medium */

#alert { color: red; }   /*
strongest */

<p id="alert"
   class="note">
  Which color wins?
</p>
```

LIVE PREVIEW

The Cascade — Specificity Hierarchy

When multiple rules target the same element, CSS resolves the conflict using this priority order, from weakest to strongest:

1. **Element selectors**
   Lowest specificity. e.g. `p { }`
2. **Class, attribute, and pseudo-class selectors**
   e.g. `.note { }`, `[type="text"] { }`, `:hover { }`
3. **ID selectors**
   e.g. `#alert { }` — beats any number of class selectors

4 **Inline styles**

e.g. `style="color:red"` directly on the tag — beats all selectors in external/internal CSS

5 `!important`

Highest possible priority — overrides EVERYTHING, including inline styles. Use only as an absolute last resort; heavy use makes CSS unmaintainable.

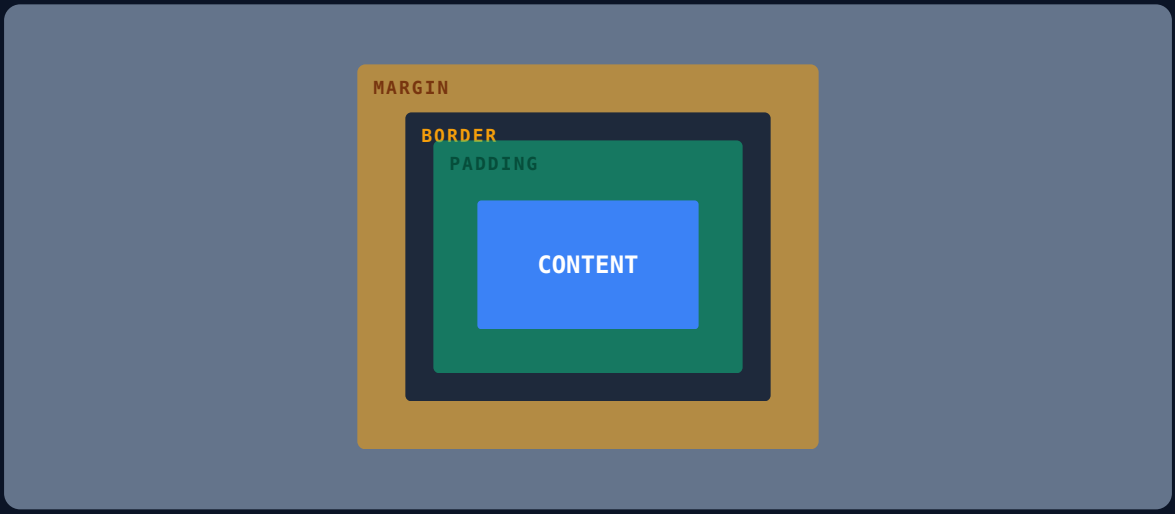
> **Gotcha Alert — When Specificity Ties**
>
> If two rules have IDENTICAL specificity (e.g. two class selectors), the one written LATER in the CSS file wins. This is the actual "cascading" behavior — order matters as a tiebreaker.

CHAPTER 06

The Box Model, Display & Positioning

Every single HTML element is, fundamentally, a rectangular box. The Box Model describes the four layers that make up that box. This is the single most exam-tested CSS concept in WS 101 — master this diagram completely.

The Box Model — Four Layers



Layer	What It Is	CSS Property
Content	The actual text, image, or content itself	`width` , `height`
Padding	Transparent space INSIDE the border, cushioning the content	`padding`
Border	A visible (or invisible) line wrapping the padding	`border`
Margin	Transparent space OUTSIDE the border, pushing away other elements	`margin`

Filipino Analogy — The Box Model

Think of a **bilao of food at a kamayan feast**. The Content is the actual rice and viands. The Padding is the banana leaf lining directly around the food. The Border is the rim of the bilao tray itself. The Margin is the empty table space between your bilao and the next person's bilao — keeping servings from touching.

box-sizing — The Property That Changes Everything

Value	Behavior
`content-box` (default)	Width/height apply ONLY to content. Padding and border are ADDED on top, making the box visually bigger than the set width.
`border-box`	Width/height include padding and border. The box stays exactly the size you set — padding/ border eat into the content space instead of expanding it.

CSS

```
* {
  box-sizing: border-box; /* near-universal best practice */
}
```

Industry Standard Practice

Almost every professional CSS file starts with `* { box-sizing: border-box; }`. It makes sizing predictable — a 300px box stays exactly 300px no matter how much padding or border you add, which matches how most developers intuitively expect sizing to work.

Display Property

Value	Behavior	Examples
`block`	Starts on a new line, fills full available width	`div, p, h1, section`
`inline`	Flows within text, only as wide as content, CANNOT set width/height/vertical margin	`span, a, strong, em`
`inline-block`	Flows within text like inline, BUT can have width/height/margin set	Common for nav links, buttons in a row
`flex`	Turns the element into a flex container — children arrange via Flexbox rules (Chapter 08)	Modern layout standard
`grid`	Turns the element into a grid container (Chapter 09)	2D layout systems
`none`	Completely removes the element from the page — takes up zero space	Hiding modals, toggling menus

Live: Block vs Inline vs Inline-Block

LIVE PREVIEW

Positioning

Value	Behavior
`static` (default)	Normal document flow. `top/left/right/bottom` have NO effect.
`relative`	Stays in normal flow, but can be nudged via `top/left/right/bottom` RELATIVE to where it would normally sit. Leaves a gap behind where it would have been.
`absolute`	Removed from normal flow entirely. Positioned relative to its nearest ancestor that has `position: relative` (or the page if none exists).
`fixed`	Removed from flow, positioned relative to the BROWSER WINDOW. Stays in place even when scrolling — used for sticky headers/ floating buttons.
`sticky`	Hybrid: acts like `relative` until the page scrolls past a threshold, then "sticks" like `fixed`.

Live: Relative + Absolute Positioning

LIVE PREVIEW

Key Rule — absolute Needs a relative Parent

An absolutely positioned element anchors itself to the nearest ANCESTOR with `position: relative` set. This is exactly how the red "NEW" badge above stays pinned to the top-right corner of its gray container, instead of the entire page.

Typography, Color & Backgrounds

This chapter covers how to control text appearance, the four ways to specify color in CSS, and how to style element backgrounds.

Typography Properties

Property	Controls	Example Values
`font-family`	Typeface	`Arial, sans-serif` — always list a fallback in case the first font is unavailable
`font-size`	Text size	`16px`, `1.2rem`
`font-weight`	Boldness	`normal` (400), `bold` (700), or numeric 100–900
`font-style`	Italics	`normal`, `italic`
`line-height`	Vertical spacing between lines of text	`1.5` (unitless multiplier is best practice)
`text-align`	Horizontal alignment	`left, center, right, justify`
`text-decoration`	Underline/strikethrough	`none, underline, line-through`
`text-transform`	Case conversion	`uppercase, lowercase, capitalize`
`letter-spacing`	Space between characters	`1px`, `0.05em`

Live: Font Family Comparison

LIVE PREVIEW

Specifying Color — Four Methods

Method	Syntax	Notes
Named Colors	`red`, `steelblue`	147 predefined keyword names. Limited precision.
Hexadecimal	`#3B82F6`	6 hex digits = RR GG BB, directly tying back to CC 101 Ch. 3 hex conversions. `#FFFFFF` = white, `#000000` = black.
RGB / RGBA	`rgb(59,130,246)` / `rgba(59,130,246,0.5)`	Red, Green, Blue values 0–255. The "A" (Alpha) controls transparency, 0 = invisible, 1 = fully opaque.
HSL	`hsl(217,91%,60%)`	Hue (0–360° on a color wheel), Saturation %, Lightness %. Most intuitive for humans to hand-tune.

Live: Same Blue, Four Ways

Named
royalblue

Hex
#3B82F6

RGB
rgb(59,130,246)

HSL
hsl(217,91%,60%)

Backgrounds

Property	Controls
`background-color`	Solid fill color
`background-image`	Image or gradient: `url('photo.jpg')` or `linear-gradient(...)`
`background-size`	`cover` (fills, may crop), `contain` (fits fully, may letterbox)

Property	Controls
`background-position`	Where the image is anchored, e.g. `center`, `top right`
`background-repeat`	`repeat` (default, tiles), `no-repeat` (shows once)

Live: Gradient Backgrounds

LIVE PREVIEW

Borders & Border Radius

CSS

```
.card {
  border: 2px solid #3B82F6;  /* width style color */
  border-radius: 12px;         /* rounded corners */
}
.circle {
  border-radius: 50%;          /* perfect circle on a square box */
}
```

CHAPTER 08

CSS Flexbox

Flexbox (Flexible Box Layout) is the modern standard for arranging elements in a single row or column, with powerful alignment and spacing control. It solved decades of layout hacks (floats, table-based layouts) and is the **priority layout system** for WS 101 — master this chapter thoroughly.

Activating Flexbox

CSS

```
.container {
  display: flex;  /* this one line turns ALL direct children into flex
items */
}
```

Once `display: flex` is set on a parent, all of its direct children automatically become "flex items" arranged in a row by default — no floats, no manual positioning needed.

flex-direction

Sets the **main axis** — the primary direction items flow along.

LIVE PREVIEW

justify-content — Main Axis Alignment

Controls spacing of items along the main axis (horizontal, for default row direction).

LIVE PREVIEW

align-items — Cross Axis Alignment

Controls alignment PERPENDICULAR to the main axis (vertical, for default row direction).

LIVE PREVIEW

flex-wrap

By default, flex items try to squeeze onto ONE line, shrinking if necessary. `flex-wrap: wrap` allows items to flow onto multiple lines instead.

LIVE PREVIEW

Flexbox Property Quick Reference

Property	Applied To	Purpose
`display: flex`	Parent	Activates flexbox for direct children
`flex-direction`	Parent	row / column / row-reverse / column-reverse
`justify-content`	Parent	Aligns items along the main axis
`align-items`	Parent	Aligns items along the cross axis
`flex-wrap`	Parent	nowrap (default) / wrap
`gap`	Parent	Sets uniform spacing BETWEEN items, no manual margins needed
`flex-grow`	Child	How much an item expands to fill leftover space (ratio-based)
`flex-shrink`	Child	How much an item shrinks when space is tight
`align-self`	Child	Overrides `align-items` for ONE specific item

Filipino Analogy — Flexbox

Think of **jeepney seating**. `flex-direction` is whether passengers sit facing each other in a row (row) or single-file (column). `justify-content` is how passengers space themselves along the bench — bunched at the front (flex-start), evenly spread with gaps (space-between). `align-items` is whether everyone aligns to the same height on the bench (stretch) or sits at varied heights (flex-start/center/flex-end).

CSS Grid, Units & Responsive Design

This final chapter covers two-dimensional layout with Grid, the measurement units CSS uses, and how to make pages adapt across phone, tablet, and desktop screens — essential since you build and test everything on your GT 30 Pro.

CSS Grid — Two-Dimensional Layout

Where Flexbox arranges items along ONE axis (row OR column), Grid arranges items along BOTH axes simultaneously — true rows AND columns at once.

CSS

```css
.gallery {
  display: grid;
  grid-template-columns: repeat(3, 1fr);  /* 3 equal columns */
  gap: 10px;
}
```

Live: 3-Column Grid Gallery

LIVE PREVIEW

Live: Grid with Spanning Items

An item can span multiple columns or rows using `grid-column` / `grid-row` — useful for hero banners or featured cards inside a grid.

CSS

```css
.featured {
  grid-column: span 2;
  /* spans 2 columns wide */
}
```

LIVE PREVIEW

Flexbox vs Grid — When to Use Which

	Flexbox	Grid
Dimensions	One-dimensional (row OR column)	Two-dimensional (rows AND columns together)
Best for	Navbars, button groups, centering content, small components	Full page layouts, photo galleries, dashboards
Content-driven?	Yes — sizing often driven by content	No — layout-driven, content fits into a predefined structure

CSS Units

Unit	Type	Relative To	Common Use
`px`	Absolute	Fixed pixel count, never changes	Borders, small precise details
`%`	Relative	Parent element's size	Fluid widths
`em`	Relative	The font-size of the CURRENT element (compounds when nested)	Padding/spacing that should scale with text
`rem`	Relative	The ROOT (`<html>`) font-size only — does not compound	Font sizes, modern best practice for consistent scaling
`vw`	Relative	1% of Viewport Width	Full-bleed hero sections, responsive type
`vh`	Relative	1% of Viewport Height	Full-screen sections (`100vh` = exactly one screen tall)

Gotcha Alert — em vs rem

`em` compounds: if a parent has `font-size: 1.2em` and a child also has `1.2em`, the child's ACTUAL size is 1.2 × 1.2 × base size — sizes spiral unpredictably when nested. `rem` always refers to the root font-size only, making it far more predictable. Most professionals default to `rem` for this reason.

Responsive Design & Media Queries

A **media query** applies CSS rules ONLY when certain conditions are met — most commonly, screen width.

CSS

```
/* Mobile-first base styles (applies to ALL screens) */
.container {
  display: flex;
  flex-direction: column;
}

/* Tablet and up */
@media (min-width: 768px) {
  .container {
    flex-direction: row;
  }
}

/* Desktop and up */
@media (min-width: 1024px) {
  .container {
    max-width: 1200px;
    margin: 0 auto;
  }
}
```

Common Breakpoints

Device Class	Typical Breakpoint
Mobile	

Device Class	Typical Breakpoint
	0 – 767px (your GT 30 Pro's browser viewport falls here)
Tablet	768px – 1023px
Desktop	1024px and above

Mobile-First vs Desktop-First

Approach	Strategy	Why It Matters
Mobile-First (recommended)	Write base CSS for small screens, then use `min-width` media queries to ADD complexity for larger screens	Forces simpler, performance-friendly design first. Matches how most users now browse — on phones.
Desktop-First	Write base CSS for large screens, then use `max-width` media queries to REMOVE/adjust for smaller screens	Older approach, often results in bloated mobile experiences carrying unnecessary desktop styling

Forward Link — HCI 101 & WS 102

Everything in this chapter is the direct foundation for **HCI 101** (Human Computer Interaction, Year 2 Sem 2, requires WS 102) where responsive, user-centered design becomes the entire focus. The JavaScript chapters that follow build interactivity directly on top of these responsive structures — and **WS 102** (Sem 2) deepens that further with more advanced client-side techniques.

Quick Reference — Building Your First Real Page

1. **Structure first**
   Write complete, semantic HTML with no styling at all. Confirm content order makes sense unstyled.
2. **Reset/normalize basics**
   Start your CSS with `* { box-sizing: border-box; margin: 0; padding: 0; }`
3. **Layout with Flexbox/Grid**
   Establish the overall page skeleton positioning before touching colors or fonts.

4. **Style details**
   Typography, colors, backgrounds, borders — the visual polish layer.

5. **Test responsiveness**
   Use VS Code Live Server and resize the browser (or open dev tools device toolbar) to check mobile/tablet/desktop breakpoints.

CHAPTER 10

JavaScript Fundamentals — Variables, Data & Strings

JavaScript is the third pillar of the web — where HTML gives structure and CSS gives appearance, JavaScript gives **behavior**. It is a **scripting language**: it runs inside the browser, reacting to what the user does and modifying the page live, without a page reload.

What JavaScript Can Do

Capability	Example
Scrolling messages	Marquee-style ticker text
Animation & dynamic images	Image sliders, hover-triggered swaps
Data input forms	Client-side validation before submitting to a server
Pop-up windows	`alert()`, `confirm()`, `prompt()`
Interactive surveys/questionnaires	Live score calculation, conditional follow-up questions

Why It's Called a "Scripting Language"

JavaScript is interpreted line-by-line by the browser at runtime, is always visible in the browser's "View Source"/DevTools, runs by default in every modern browser, and behaves identically across platforms. Your lecture deck lists this combination as JS's "unique features": script-based output, code visibility, full functionality, simplicity, default scripting support, and cross-platform support.

Variables

A variable is a named container for a value. The lecture deck declares variables with `var` — the original JavaScript keyword. Modern code favors `let` / `const`, which behave similarly but scope more predictably; both are worth recognizing.

Naming Conventions

Valid	Invalid	Why
`idnumber`	`id number`	No spaces allowed in a variable name
`Customer_ID`	`Customer.ID`	No periods — use underscores instead
`SY2016`	`2016`	Cannot start with a digit
`_Continue`	`Continue`	Reserved keywords need a prefix — `continue` is a JS keyword

Global vs Local Variables

Type	Scope
Global Variable	Declared outside any function — accessible anywhere in the script
Local Variable	Declared inside a function — only accessible within that function

Data Types

Type	Example
Alphabetical	`"A"`
Integer / Floating-point	`10` , `3.14`
Boolean / Logical	`true` , `false`
String	`"Welcome to QCU"`
Null value	`null` — deliberately empty

Form input often arrives as text even when it looks numeric. Two built-in functions convert it:

Function	Converts To	Example
`parseInt()`	Whole number	`parseInt("25")` → `25`
`parseFloat()`	Decimal number	`parseFloat("3.14")` → `3.14`

Live: Variables, Data Types & Conversion

HTML

```
<script>
var schoolName = "QCU";
var unitsText = "3";
var units = parseInt(unitsText);
var isEnrolled = true;

document.write(
  schoolName + " - " + units +
  " units - Enrolled: " +
isEnrolled
```

LIVE PREVIEW

```
);
</script>
```

Literals

A **literal** is a fixed value written directly into code, as opposed to a variable that can change.

Type	Example
Numeric Literal	`var width = 3;`
String Literal	`var schoolName = "QCU";`

Strings

A string can be created as a simple value, or explicitly as a String object with `new`:

JS

```
strTest = "Welcome to QCU";
// or, as an object:
strTest = new String("Welcome to QCU");
```

Live: String Concatenation

HTML

```
<script>
var strTest1 = "Welcome to QCU.";
var strTest2 = "Enjoy your
stay.";
var strBoth = strTest1 + " " +
strTest2;
```

LIVE PREVIEW

```
document.write(strBoth);
</script>
```

Useful String Methods

Method/Property	Effect	Example
`.length`	Property — counts characters	`"Welcome to QCU".length` → `14`
`.toUpperCase()`	Converts to ALL CAPS	`strTest.toUpperCase()`
`.toLowerCase()`	Converts to all lowercase	`strTest.toLowerCase()`

Expressions & Operators

Operator	Meaning	Example
`+`	Addition (numbers) or concatenation (strings)	`result = 5 + 7;` → `12`
`-`	Subtraction	`score = score - 1;`
`*`	Multiplication	`total = quantity * price;`
`/`	Division	`average = sum / 4;`
`%`	Modulo — remainder after division	`remainder = sum % 4;`
`++`	Increment by 1	`tries++;`
`--`	Decrement by 1	`total--;`

Arrays

An array is a special variable that holds MULTIPLE values under one name, accessed by a numeric index starting at `0`.

JS

```
var score = new Array();
score[0] = 39;
score[1] = 40;
score[2] = 100;
score[3] = 49;

// modern shorthand — same
result:
var score2 = [39, 40, 100, 49];
```

LIVE PREVIEW

Forward Link — CC 102

Arrays, variables, and data types here are the same concepts you're building in **CC 102** (Java) — just different syntax. A JavaScript array and a Java array both use zero-based indexing and store an ordered collection under one name.

CHAPTER 11

Control Structures, Functions & Objects

With data covered, this chapter covers how JavaScript makes DECISIONS (conditionals), REPEATS actions (loops), REUSES code (functions), and ORGANIZES related data (objects).

Flow of Control & Flowcharts

A **flowchart** diagrams the order operations execute in. Ovals mark start/end, parallelograms mark input/output, and diamonds mark decision points with Yes/No branches.

Deck Example

Lecture deck's flowchart: Leave home → Check time → (diamond) Before 7am? → Yes: Take bus / No: Take subway → Reach school. This is exactly the logic an `if/else` statement encodes below.

Conditional Statements — The If Statement

JS

```
var x = 10;
if (x == 10) {
  window.alert("The number is 10!");
}
```

Conditional Expressions

Operator	Meaning	Example
`==`	Equal to (value only, no type check)	`x == 10`
`===`	Strictly equal (value AND type) — modern best practice	`x === 10`
`<` / `>`	Less than / greater than	`y < z`
`&&`	Logical AND — both sides must be true	`x > 0 && x < 10`
`||`	Logical OR — at least one side true	`x == 1 || x == 2`

Live: If/Else Grade Checker

HTML

```
<script>
var grade = 78;
if (grade >= 75) {
```

LIVE PREVIEW

```
  document.write("PASSED");
} else {
  document.write("FAILED");
}
</script>
```

Looping Structures

Loop Type	Checks Condition	Best For
`for`	Before each iteration	A known, fixed number of repetitions
`while`	Before each iteration	Unknown repetitions, condition-driven
`do...while`	AFTER each iteration — always runs at least once	When the loop body must run at least one time regardless

JS

```
// FOR LOOP
for (i = 0; i < 10; i++) {
  document.write("this is line "
+ i + "<br>");
}

// WHILE LOOP
while (total < 10) {
  ctr++;
  total += values[ctr];
}

// DO-WHILE LOOP
do {
  n++;
  total += values[n];
} while (total < 10);
```

LIVE PREVIEW — FOR LOOP

Functions

A function is a named, reusable block of code. It runs only when CALLED, and can accept input values called **arguments**.

JS

```
function Greet() {
  alert("Hello World!");
}

// with an argument:
function Greet(who) {
  alert("Greetings, " + who);
}

Greet("Kian");   // -> "Greetings, Kian"
```

Objects — Properties & Methods

An **object** bundles related data (**properties**) and actions (**methods**) under one name, accessed with dot notation.

Concept	Example	Meaning
Property	`txtGreet.length`	A stored value belonging to the object — no `()`
Method	`txtGreet.indexOf()`	An action the object can perform — always called with `()`

Creating a Custom Object

JS

```
var person = new Object();
person.firstname = "Juan";
person.lastname = "Santos";
person.age = 19;
```

```
var x = person.age;  // x now holds 19
```

Built-In Objects — String Objects

```
JS
var string1 = new Object();
string1.prop = "I am a property of string1 object";
alert(string1.prop);
```

> **Forward Link — PF 101**
>
> Objects with properties and methods here are your first exposure to **Object-Oriented Programming** — the entire subject of **PF 101** (Year 2). A JS object's "properties" map to a Java class's "fields," and its "methods" map directly to Java class methods.

CHAPTER 12

Regular Expressions, Events & Embedding JavaScript

This closing JS chapter covers pattern-matching text with regular expressions, making pages react to user actions with events, and the different ways JavaScript code actually attaches to an HTML page.

Regular Expressions (Regex)

A regular expression is a search PATTERN used to match, find, or replace text.

```
JS
// object form:
var regexPatt = new RegExp(pattern, modifiers);
```

```
// shorthand/literal form (more common):
var regexPatt = /pattern/modifiers;
```

Modifiers

Modifier	Effect
`i`	Case-insensitive matching
`g`	Global matching — finds ALL matches, not just the first
`m`	Multiline matching

Patterns

Pattern	Type	Matches
`/abc/`	Simple pattern	The literal characters "abc" anywhere in a string
`/ab*c/`	Special character (`*` = zero or more)	"ac", "abc", "abbbbc" — any number of "b"s between a and c

Pattern Matching Methods

Method	Belongs To	Returns
`.match()`	String	Array of matches
`.search()`	String	Index position of first match
`.replace()`	String	New string with matches replaced
`.split()`	String	Array split at each match
`.exec()`	RegExp object	Match details, or `null`
`.test()`	RegExp object	`true` / `false` — does it match at all

JS

```
var str = "The rain in Spain stays mainly in the plain";
var res = str.match(/ain/g);
// res -> ["ain", "ain", "ain"]  (from rain, mainly, plain)
```

RegExp Object Reference

Category	Members
Methods	`exec()`, `test()`
Properties	`source`, `global`, `ignoreCase`, `multiline`

Events & Event Handlers

Event Category	Examples
Mouse events	`onclick`, `onmouseover`, `onmouseout`
Keyboard events	`onkeydown`, `onkeyup`
Frame/Object events	`onload`, `onresize`
Form events	`onsubmit`, `onchange`, `onfocus`

Live: Event Handler in Action

HTML

```
<button
onclick="document.write(Date())">
  Check Time
</button>
```

LIVE PREVIEW — TAPPABLE

Gotcha — alert() in Sandboxed Previews

Popup dialogs like `alert()`, `confirm()`, and `prompt()` are often blocked inside sandboxed iframes (like this guide's live previews) for security reasons — but work normally in a real browser tab or VS Code Live Server. Also note: calling `document.write()` from an event handler AFTER the page has loaded erases the whole page rather than appending to it — that's why the live demo above updates an element's `innerHTML` instead, which is the safer real-world pattern.

Popup Boxes

Type	Function	Returns
Alert Box	`alert("message")`	Nothing — just displays and waits for OK
Confirm Box `supplementary`	`confirm("message")`	`true` (OK) or `false` (Cancel)
Prompt Box `supplementary`	`prompt("message", "default")`	The text typed, or `null` if cancelled

Confirm and Prompt aren't on the lecture slides, but they're the other two standard JS popup types alongside Alert, and follow the exact same calling pattern — worth knowing as a set.

Client-Side JavaScript — Core Objects

Object	Represents
`window`	The entire browser window/tab
`location`	The current page's URL
`document`	The current page's content (the DOM)
Element objects	Individual tags on the page (a specific button, input, div)

Embedding JavaScript in HTML — 4 Ways

Method	Example
1. Inline scripting	`<script> /* code */ </script>`
2. External scripting	`<script src="scriptLocation.js"></script>`
3. Via HTML event handler attribute	`<input onchange="order.giftwrap = this.checked;">`
4. Via `javascript:` protocol	`<a href="javascript:alert('Hi')">Click</a>`

Best Practice

Method 2 (external `.js` file) is standard for real projects — same reasoning as external CSS: separation of concerns, reusability across pages, and browser caching. Methods 3 and 4 mix JS directly into HTML attributes and are considered messy by modern standards, even though they still work and appear on your slides.

Where This Guide Goes Next

HTML (Weeks 2–3) + CSS (Week 4) + JavaScript (Weeks 5–6) cover the front-end half of WS 101 and are the midterm material. The course then moves to server-side work for finals: relational database design and SQL (Weeks 10–11, covered in Chapters 13–14) and the Laravel framework (Weeks 12–14, covered in Chapters 15–16). Weeks 7–9 are the midterm exam period and aren't part of either source deck.

Relational Model, ER Diagrams & UML

Weeks 10–11 shift WS 101 from front-end code to **database design** — how data gets organized before any PHP or SQL touches it. This chapter covers the theory: hierarchical structures, the relational model, ER diagrams, and UML. Chapter 14 covers writing the actual SQL and connecting PHP to it.

Hierarchical Data Structures

Before relational databases became the standard, data was modeled as a **hierarchical structure** — an upside-down tree showing parent-child relationships. It's composed of **segments** (the equivalent of a file system's record type), each of which can have a **parent** and **children**. The topmost segment is the **root segment**, with everything below it organized into levels (Level 1, Level 2, Level 3...).

Term	Meaning
Hierarchy chart	A diagram created to document a program or database — conveys the big picture of the modules used, or the information a database contains
Hierarchical path	An ordered sequencing of segments tracing the hierarchical structure
Preorder traversal	Also called the *hierarchical sequence* or *left-list path* — traces every segment starting from the root, beginning at the leftmost segment first

Why This Still Matters

Even though most modern databases are relational (tables, not trees), hierarchy charts and left-to-right traversal thinking still show up constantly — file system folders, HTML's DOM tree, JSON nesting, and org charts all follow the same parent → children logic.

The Relational Model

Dr. E.F. Codd, a British scientist working for IBM, invented the **relational model** for database management. Instead of trees, it represents a database as a collection of **relations** — a formal name for **tables** made up of rows and columns.

Relational Term	Everyday Name	Example
Relation	Table	`Employee`
Attribute	Column / field	`Emp_Name` , `Emp_Age`
Tuple	Row / record	One employee's full data
Primary Key	Unique row identifier	`Emp_ID`

Emp_ID (PK)	Emp_Name	Emp_Address	Emp_Age
00001	Melanie	Novaliches	27
00002	Michael	Cainta	26
00003	Phoebe	Marikina	20
00004	Lomel	Taguig	28

Relating Tables — Foreign Keys

A single table can only hold so much before it gets messy. Relational design splits data across multiple tables, then reconnects them using a **foreign key** — a column that references another table's primary key. A *Sales-summary* table can link an `Employee` table to an `Items` table without duplicating employee or item details inside it.

Table	Role	Key Columns
Employee	Stores employee records	`Emp_ID` (PK)

Table	Role	Key Columns
Items	Stores item records	`Item_ID` (PK)
Sales-summary	Links employees to the items they sold	`Emp_ID`, `Item_ID` (both FKs)

The Payoff

If Melanie's address changes, you update *one* row in `Employee` — every sale record referencing her `Emp_ID` stays accurate automatically. That's the core reason relational design beats one giant flat table.

Entity-Relationship (ER) Modeling

Before building tables, designers sketch an **Entity-Relationship (ER) Diagram** — a graphical way to express a database's overall logical structure. The classic teaching example: two **PERSON** entities playing the **roles** of **Husband** and **Wife**, connected by the relationship **"is married to."** Nouns become entities; verbs become relationships.

ER Diagram Notation

Symbol	Component	Meaning
Rectangle	Entity	A distinct object or thing (a person, a course, a product)
Double rectangle	Weak entity	An entity that can't be uniquely identified without another entity it depends on
Underlined oval	Key attribute	The attribute that uniquely identifies each instance (e.g. `StudentNo`)
Double oval	Multivalued attribute	An attribute that can hold more than one value at once
Dashed oval	Derived attribute	A value calculated from other data rather than

Symbol	Component	Meaning
		stored directly (e.g. Age from Birthdate)
Diamond	Relationship	How two entities connect (e.g. *studies*, *enrolls in*)

Cardinality — Crow's Foot Notation

Cardinality describes how many instances of one entity can relate to instances of another. The most common notation style is **crow's foot**, drawn at the end of the connecting line closest to the entity it describes.

Notation	Meaning
One to one	Exactly one record relates to exactly one record on the other side
One to many (mandatory)	One record relates to at least one, possibly many, records on the other side
Many	Multiple records on this side can relate to the other entity
One or more (mandatory)	At least one relationship must exist
One and only one (mandatory)	Exactly one relationship is required — no more, no fewer
Zero or one (optional)	The relationship may or may not exist, but never more than one
Zero or many (optional)	The relationship is fully optional, and can repeat

Reading a Diagram — Worked Examples

1. **Student — Class**
   A many-to-many relationship: a Student can take many Classes, and a Class can hold many Students.

2. **Student — Class — Instructor**
   Adds a third entity: each Class also connects to one Instructor, forming a chain rather than a single pair.

3. **Student — Subject**
   Many-to-many, crow's feet on both ends: Students study many Subjects, and each Subject is studied by many Students.

4. **Student entity with attributes**
   `StudentNo` (underlined = key attribute), `StudName`, `Address`, `ContactNo` attach directly to the Student rectangle.

5. **Student *studies* Course — the full pattern**
   Student (`StudentNo`, `StudName`, `Address`, `ContactNo`) connects through the *studies* relationship diamond to Course (`CourseCode`, `CourseTitle`) — two entities, their attributes, and the relationship joining them.

Unified Modeling Language (UML)

UML is a broader modeling approach than ER diagrams — it models many aspects of software development for object-oriented systems, not just the database layer.

Diagram Type	What It Shows
Class	A static structure diagram — a system's classes, their attributes, methods, and the relationships between classes
Use Case	How the system behaves from the end user's point of view
State	The possible states an object can be in
Sequence	The sequence of interactions between objects over time
Activity	The business and operational step-by-step workflow of a system's components
Component	How a system's physical components are organized and connected

ER Diagram vs. Class Diagram — Don't Mix Them Up

Both use rectangles and lines, which makes them easy to confuse. An **ER diagram** models *data* — what tables and columns exist and how they relate. A UML **Class diagram** models *code* — what classes, methods, and objects exist and how they relate. ER comes first, when you're designing the database; Class diagrams come later, when you're designing the application logic that will use it.

CHAPTER 14

SQL & Connecting PHP to MySQL

Structured Query Language — Three Components

SQL (Structured Query Language) is how you talk to a relational database. It splits into three sub-languages, each with a distinct job:

Component	Full Name	Job
DML `most used`	Data Manipulation Language	Retrieve, update, add, or delete data *inside* a database (`SELECT` , `INSERT` , `UPDATE` , `DELETE`)
DDL	Data Definition Language	Create and modify the database structure itself (`CREATE` , `ALTER` , `DROP`)
DCL	Data Control Language	Maintain proper security and permissions for the database (`GRANT` , `REVOKE`)

Data Definition — Creating & Modifying Structure

SQL

```sql
-- create and delete a database
CREATE DATABASE DBEnrollment;
DROP DATABASE DBEnrollment;

-- create a table
CREATE TABLE tblStudent (
  studentNumber INT PRIMARY KEY,
  studentName   VARCHAR(50),
  Course        VARCHAR(20)
);

-- add a column to an existing table
ALTER TABLE tblStudent ADD section VARCHAR(10);

-- remove a column from an existing table
ALTER TABLE tblStudent DROP COLUMN Course;

-- delete an entire table
DROP TABLE tblStudent;
```

Basic Data Retrieval — SELECT

Three clauses do almost all the work of reading data back out of a table:

Clause	Job
`SELECT`	Lists which attributes (columns) you want in the result
`FROM`	Lists which table(s) to scan
`WHERE`	Filters rows using a condition on the attributes named in `FROM`

SQL

```sql
SELECT column_name FROM table_name;

-- or select every column:
SELECT * FROM table_name;
```

Modifying Table Data — INSERT, UPDATE, DELETE

Statement	Job
`INSERT`	Adds new rows into a table
`UPDATE`	Edits existing data in specific rows and columns
`DELETE`	Removes entire rows

SQL

```sql
INSERT INTO tblStudent (LastName, FirstName, CourseCode)
VALUES ('Reyes', 'Melanie', 'BSIT');

UPDATE tblStudent SET CourseCode = 'BSN' WHERE StudentNumber = '1069';

DELETE FROM tblStudent WHERE StudentNumber = '1000';
```

Fixing a Slide Typo — Multi-Row INSERT

The lecture slide's INSERT example crams two students into one `VALUES()` clause with 3 columns but 6 values — that's invalid SQL. To insert Reyes/Melanie *and* Smith/John in one statement, use two separate parenthesized groups instead:

```sql
INSERT INTO tblStudent (LastName, FirstName, CourseCode)
VALUES ('Reyes', 'Melanie', 'BSIT'),
       ('Smith', 'John', 'BSED');
```

MySQL Data Types

Type	Description
`INT`	Whole numbers, roughly -2,147,483,648 to 2,147,483,647
`DECIMAL`	Floating-point / decimal numbers
`CHAR`	Fixed-length string, max 255 characters
`VARCHAR`	Variable-length string, max 255 characters
`TEXT`	Long string, max 65,535 characters
`DATE`	`YYYY-MM-DD`
`TIME`	`HH:MM:SS`
`DATETIME`	`YYYY-MM-DD HH:MM:SS` combined

Connecting PHP to MySQL

PHP talks to MySQL through the `mysqli_connect()` function, which needs four pieces of information:

Parameter	Description
Host	A hostname or IP address — usually `"localhost"` in development
Username	The MySQL account username — often `"root"` on a local Xampp install
Password	The MySQL account password
Database	The name of the specific database to use

PHP

```php
<?php
$dbServername = "localhost";
$dbUsername   = "root";
$dbPassword   = "";
$dbName       = "qcu";

$conn = mysqli_connect($dbServername, $dbUsername, $dbPassword, $dbName);
?>
```

Centralizing the Connection

Rather than repeating the connection code on every page, save it once (commonly as `includes/dbh.php`, short for "database handler") and pull it into any page that needs it:

PHP

```php
<?php
include_once 'includes/dbh.php';
?>
<!DOCTYPE html>
<html>
<head><title></title></head>
...
```

Why include_once, Not include

`include_once` guarantees the file is only pulled in once, even if several other included files also try to include it — this prevents "function already defined" fatal errors from a connection script loading twice.

Reading Data — SELECT via PHP

PHP

```php
<?php
include_once 'includes/db.php';
?>
<!DOCTYPE html>
```

```
<html>
<head><title></title></head>
<body>

<?php
  $sql = "SELECT * FROM stud;";
  $result = mysqli_query($conn, $sql);
  $resultCheck = mysqli_num_rows($result);

  if ($resultCheck > 0) {
    while ($row = mysqli_fetch_assoc($result)) {
      echo $row['stud_uid'] . "<br>";
    }
  }
?>

</body>
</html>
```

1. **mysqli_query($conn, $sql)**
   Sends the SQL string to the database over the connection and returns a result set.

2. **mysqli_num_rows($result)**
   Counts how many rows came back — used here to check "did we actually get anything?" before looping.

3. **mysqli_fetch_assoc($result)**
   Pulls one row at a time as an associative array (`$row['column_name']`). The `while` loop keeps calling it until no rows are left, at which point it returns `false` and the loop ends.

Writing Data — INSERT via PHP

PHP

```
<body>
<?php
  $sql = "INSERT INTO stud (stud_FirstName, stud_LastName, stud_email,
stud_uid, stud_pwd)
          VALUES ('Paolo', 'Lopez', 'orven@gmail.com', 'Paolo321',
'Lopez123');";
  $result = mysqli_query($conn, $sql);
```

```
?>
</body>
```

Beyond the Slides — SQL Injection

Building queries by concatenating raw variables straight into the SQL string, as shown above, is exactly how SQL injection attacks work — if `$sql` ever includes unfiltered user input (like a login form field) instead of hardcoded values, an attacker can inject their own SQL. Production code uses **prepared statements** (`mysqli_prepare()` with `?` placeholders, or PDO) instead. It's not on this deck, but it's the single most important upgrade to make once you're past the fundamentals.

CHAPTER 15

Laravel Fundamentals — Installation, Structure & Routing

Weeks 12–14 introduce **Laravel**, a PHP framework that automates much of the plumbing raw PHP+MySQL requires — routing, templating, and database schema management. This chapter covers install and routing (Week 12); Chapter 16 covers controllers, Blade views, and database migrations (Weeks 13–14).

What Is Laravel?

Laravel is a PHP framework — a large, pre-built collection of tools and conventions that handles the repetitive plumbing of a web application so you write far less code than raw PHP requires.

Benefit	Why It Matters
Quick and simple	Built-in tools (routing, ORM, templating) replace hundreds of lines of manual PHP
Security	

Benefit	Why It Matters
	Built-in protection against common attacks, including SQL injection and CSRF
Better performance	Caching and optimization tools ship with the framework
Traffic handling	Designed to scale for high-traffic applications
Flexible	Not opinionated about how you structure business logic
Third-party integrations	Straightforward to plug in payment gateways, mail services, etc.
Simple maintenance	Consistent structure makes it easier for other developers to pick up your code
Cost	Free and open-source — no licensing fees

Installing Laravel — Prerequisites

1. **Xampp Server (5.0+)**
   Provides Apache (web server), MySQL (database), and PHP in one local package.
2. **PHP (5.5+)**
   Bundled with Xampp — Laravel needs a minimum PHP version to run.
3. **Composer**
   PHP's package manager, downloaded from `getcomposer.org`. During setup, point it to your Xampp PHP install (e.g. `xampp/php/php.exe`).

Creating a Laravel Project

1. **Open a terminal and verify Composer**
   Type `composer` and press Enter — a working install prints its version and a list of commands.
2. **Navigate to htdocs**
   Xampp serves web files from its `htdocs` folder — change into it, e.g. `cd xampp/htdocs` (adjust the drive/path to match your install).

3 **Create the project**

Composer downloads Laravel and scaffolds a new project folder here.

4 **Open it in the browser**

Visit `https://localhost/project_name/public` — Laravel's default welcome page confirms a working install.

> **Legacy Version Note**
>
> This deck's screenshots show **Laravel 5** (circa 2016). Current Laravel installs differently (`composer create-project laravel/laravel project-name` , or the `laravel new` CLI installer) and typically serves via `php artisan serve` rather than a manually configured Xampp `htdocs` folder. The *concepts* in this and the next chapter — routing, controllers, Blade, migrations — are still accurate; only a handful of install commands and filenames have moved since Laravel 5.

Routing — The Static get() Method

Routing is the concept of mapping a URI (like `/hello`) to a destination — a page, a piece of logic, or a controller method. In Laravel 5, routes were defined in `app/Http/routes.php` :

```php
Route::get('/', function () {
    return view('welcome');
});
```

`Route` is a class with a static method `get()` that takes a URI and a closure (an anonymous function). The closure runs `view('welcome')` , which returns the contents of `welcome.blade.php` — what a visitor sees when they load the site's home page.

Adding a New Route

Any new page starts the same way — call `Route::get()` with a new URI and a closure that returns something to display:

PHP

```php
Route::get('/hello', function () {
    echo "<html><head><title>Laravel</title>
    <style>
      body { font-family: 'Lato'; text-align: center; }
      .title { font-size: 110px; }
    </style></head>
    <body>
      <div class='title'>Hello, Dear Reader</div>
      <p>How are you? I hope you would love this book!</p>
    </body></html>";
});
```

Visiting `http://localhost:8000/hello` now renders the echoed HTML directly. This works for a quick demo, but hand-writing HTML inside a PHP closure doesn't scale — that's exactly the problem Blade views (Chapter 16) solve.

CHAPTER 16

Laravel MVC — Controllers, Views/ Blade & Database Migration

The Controller's Role in MVC

Laravel follows the **MVC (Model–View–Controller)** pattern. The **controller** sits in the middle — it controls the flow between the Model (data) and the View (presentation). A controller is best thought of as a **transporter**: it executes logic and returns a view, but never needs to know what that view contains or how it's displayed.

Creating a Controller

Controllers live in `app/Http/Controllers`. Here's a minimal one that just returns a view:

PHP

```php
<?php
// app/Http/Controllers/MyController.php
namespace App\Http\Controllers;

use Illuminate\Http\Request;
use App\Http\Requests;

class MyController extends Controller
{
    // returning a simple page using view
    public function returningASimplePage()
    {
        return view('newpage');
    }
}
```

Routing to a Controller Method

Once a controller exists, the route no longer holds a closure — it points to the controller and the method:

PHP

```php
// routes.php
Route::get('newpage', 'MyController@returningASimplePage');
```

Visiting `http://localhost:8000/newpage` now runs `MyController`'s `returningASimplePage()` method, which returns the `newpage` view.

> **Legacy Syntax — 'Controller@method'**
>
> The `'MyController@returningASimplePage'` string syntax was removed in Laravel 8. Current Laravel uses an array-callable form instead: `Route::get('newpage', [MyController::class, 'returningASimplePage']);`. Use the array form on a modern Laravel version even though these slides show the older string form.

Controller Rules to Remember

1. **Always use public functions**
   Laravel's router can only call methods it can access from outside the class — `public function index() { return view('home'); }`

2. **Route the controller, not a closure**
   Once a controller exists for a route, stop using inline anonymous functions — route to the controller method instead: `Route::get('home', 'MyController@index');`

Views & Blade — The Master Layout

Laravel's templating engine is called **Blade**. Rather than repeating the same HTML skeleton on every page, you build one **master layout** with `@yield()` placeholders that child views fill in:

BLADE

```
{{-- resources/views/pages/master.blade.php --}}
<!DOCTYPE html>
<html>
<head>
  @yield('head')
  <link rel='stylesheet' href='/css/style.css'>
  <title>@yield('title')</title>
</head>
<body>
  <div class="container">
    <div class="heading">@yield('heading')</div>
    <div class="content">@yield('content')</div>
    <div class="footer">@yield('footer')</div>
  </div>
</body>
</html>
```

Child Views — @extends & @section

A child view declares which master it extends, then fills each named section between `@section('name')` and `@stop`:

BLADE

```blade
{{-- resources/views/pages/about.blade.php --}}
@extends('pages.master')

@section('title')
  It's a test page
@stop

@section('heading')
  {{ $name }} {{ $profession }}
@stop

@section('content')
  <div class='h1'>Every Writing is a Problem!</div>
  <p>There was a problem few minutes back...</p>
@stop

@section('footer')
  Home of Sanjib Sinha
@stop
```

The master page plays the role of a **parent**; the child view (`about.blade.php`) fills in its blanks. Logic flows down from parent to child.

Blade Echo Syntax — Double Curly Braces

The `{{ $variable }}` syntax is Blade's shorthand for `echo`. It's not just cosmetic — Blade automatically runs the output through `htmlspecialchars()`, escaping any HTML/script characters before printing them. That's a built-in defense against cross-site scripting (XSS) that raw `echo $variable;` doesn't give you for free.

Passing Data — Controller to View

Three equivalent ways to hand variables from a controller method to its view:

PHP

```php
// app/Http/Controllers/MyController.php

// Method 1 -- compact()
public function about() {
    $name = 'John Doe';
    $profession = 'A Writer';
    return view('pages.about', compact('name', 'profession'));
}

// Method 2 -- chained with()
public function about() {
    $name = 'John Doe';
    return view('pages.about')->with('name', $name);
}

// Method 3 -- array with([...])
public function about() {
    return view('pages.about')->with([
        'name' => 'John Doe',
        'profession' => 'A Writer'
    ]);
}
```

compact() Explained

`compact('name', 'profession')` automatically builds `['name' => $name, 'profession' => $profession]` from variables already in scope — a shortcut for Method 3 when your array keys match your variable names exactly.

The .env File & Environment Config

Database credentials live in a project-root `.env` file, referenced by `config/database.php`:

ENV

```
# .env
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=testdb
```

```
DB_USERNAME=root
DB_PASSWORD=pass
```

Keeping Secrets Safe — .gitignore

If this project is ever pushed to GitHub or another cloud repo, the `.env` file (and other sensitive files) should never go with it. Laravel ships with them pre-listed in `.gitignore`:

GITIGNORE

```
# .gitignore
/vendor
/node_modules
/public/storage
Homestead.yaml
Homestead.json
.env
```

> **Double-Check Before Every Push**
>
> A committed `.env` file exposes real database passwords in your repo's history — deleting it in a later commit doesn't remove it from history. Confirm `.env` is listed in `.gitignore` *before* your first commit, not after.

Database Migration — What & Why

Before migrations, creating a database, table, and columns meant hand-writing SQL or clicking through phpMyAdmin — repetitive and hard to keep in sync across a team. **Migrations** let you define a schema as PHP code, version it alongside the rest of your project, and run it with one command.

Anatomy of a Migration File

Laravel ships with a migration that creates the default `users` table — a good reference for the pattern:

PHP

```php
<?php
// database/migrations/..._create_users_table.php
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Database\Migrations\Migration;

class CreateUsersTable extends Migration
{
    public function up()
    {
        Schema::create('users', function (Blueprint $table) {
            $table->increments('id');
            $table->string('name');
            $table->string('email')->unique();
            $table->string('password');
            $table->rememberToken();
            $table->timestamps();
        });
    }

    public function down()
    {
        Schema::drop('users');
    }
}
```

Piece	Job
`up()`	Runs when the migration is applied — defines what to build
`down()`	Runs when the migration is rolled back — undoes what `up()` did
`Schema::create()`	Creates a new table using a `Blueprint`
`$table->increments('id')`	Auto-incrementing primary key column
`$table->timestamps()`	Adds `created_at` and `updated_at` columns automatically

Migration Workflow — Create, Run, Update

1. **Generate a migration file**

   `php artisan make:migration create_tasks_table --create="tasks"` — scaffolds a new timestamped file in `database/migrations`.

2. **Define the columns inside up()**

   Edit the generated file — e.g. add `$table->string('title'); $table->text('body');` inside `Schema::create()`.

3. **Run the migration**

   `php artisan migrate` — applies every pending migration and creates the actual MySQL tables.

4. **Verify in phpMyAdmin**

   The new table (and its columns) now appears in the actual database — no manual SQL was typed.

Adding a Column After the Fact

Forgot a column after the table's already built? Don't edit the old migration — generate a new one that targets the existing table with `--table` instead of `--create`:

BASH

```
php artisan make:migration add_reminder_to_tasks_table --table="tasks"
```

Then add `$table->text('reminder')->nullable();` inside its `up()`, and run `php artisan migrate` again — Laravel only applies migrations it hasn't run yet, so this adds the new column without touching or re-creating the rest of the table.

In Summary

Database migration is a technique to create, update, or delete a table and its columns using version-controlled PHP files instead of raw SQL or a GUI tool. On a team working from different machines, everyone runs the same migration files and ends up with an identical database schema — no more "it works on my database" bugs.

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